



String Slice & Tweet Counter

Brief Overview

This note covers [JavaScript String Operations](#) and was created from an [5-page PDF](#). It provides detailed explanations of the slice method, examples of zero-based indexing, and a hands-on [Twitter character counter exercise](#) to enforce the 140-character limit.

The notes cover syntax & parameters, zero-based indexing, real-world examples, and a prompt-slice-alert flow.

Key Points

- Understand how `slice(start, end)` extracts a substring, with the end index being **exclusive**.
- Apply zero-based indexing to precisely select characters in a string.
- Use `slice(0, 140)` in a prompt-alert chain to truncate tweets to 140 characters in a single line.
- Optionally verify the final length with the `.length` property.



String slice Method



Syntax & Parameters

Definition: `slice(startIndex, endIndex)` returns a new string containing characters from `startIndex` up to, but **not including**, `endIndex`.

```
let result = str.slice(startIndex, endIndex);
```

- **startIndex** – position of the first character to include (zero-based).
- **endIndex** – position *after* the last character to include.
- If `endIndex` is omitted, the slice runs to the end of the string.



Zero-Based Indexing

- JavaScript counts characters from **0**.
- The first character is at index **0**; the *n*th character is at index ***n*-1**.

◆ Examples

Example Code	start	end	Output
name.slice(0,1)	0	1	"A"
name.slice(5,6)	5	6	"a"
name.slice(0,3)	0	3	"Ang"
name.slice(1,5)	1	5	"ngel"
abc.slice(1,3)	1	3	"bc"

- The number of characters returned equals **endIndex - startIndex**.
- Example: slice(0,3) → 3 - 0 = 3 characters.

🐦 Twitter Character Counter Exercise

◆ Goal

Enforce the 140-character limit for a tweet by **automatically truncating** any excess characters.

◆ Implementation Steps

1. Prompt the user for a tweet (prompt).
2. Use **slice(0, 140)** to keep only the first 140 characters.
3. Show the truncated tweet with alert.
4. (Optional) Replace intermediate variables with a single expression.

◆ Sample Code

```
// Prompt the user for a tweet
let tweet = prompt("Enter your tweet:");

// Truncate to the first 140 characters
let tweetUnder140 = tweet.slice(0, 140);

// Show the result
alert(tweetUnder140);
```

One-Liner Variant

```
alert(prompt("Enter your tweet:").slice(0, 140));
```

◆ Key Points

- Variable names **may contain numbers** but cannot start with a number (tweetUnder140 is valid).
 - The .length property can be used to verify the final length if desired.
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Key Takeaways

- **slice** extracts a substring using **zero-based** start and end indices.
- The end index is **exclusive**; the slice length is end - start.
- To enforce a character limit, combine prompt, slice, and alert.
- Intermediate variables are optional; a single-line expression works as long as syntax is correct.